

Pre-work · Joints

Articulations and Joints. Read the Part A chapter first. Every joint on your list can be asked about four ways, so practice answering all four every time.

Module 2 · 4 of 4

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the joints chapter into mini visuals. Seven chunks. The two classification grids are the spine of the whole chapter.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 GRID

Structure crossed with function

Fibrous, cartilaginous, synovial down the side. Cavity present, tissue holding the bones, and usual functional class across the top. This one grid answers most exam questions in this chapter.

2 HUB

Fibrous joints

Fibrous at the hub. Spoke to suture, syndesmosis, gomphosis, interosseous membrane, synostosis. One example on each spoke.

3 SPLIT

Synchondrosis against symphysis

One split. Cartilage type, mobility, and two examples on each branch. The cartilage type is what decides everything else.

4 HUB

A synovial joint in section

Joint cavity at the hub. Spoke to articular cartilage, capsule with its two membranes, ligaments, synovial fluid, fat pads, menisci, labrum, bursae.

5 GRID

The six synovial types

Type down the side. Surface shape, axes, and one example across the top. Six rows, and the axes column is what people lose.

6 GRID

The movements

Two grids. Angular movements in one, special movements in the other. Draw a tiny arrow showing each direction rather than writing a definition.

7 SPLIT

Osteoarthritis against rheumatoid arthritis

One split. Cause, which joints, and whether it is symmetrical. Three branches each. This pair appears on every exam.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

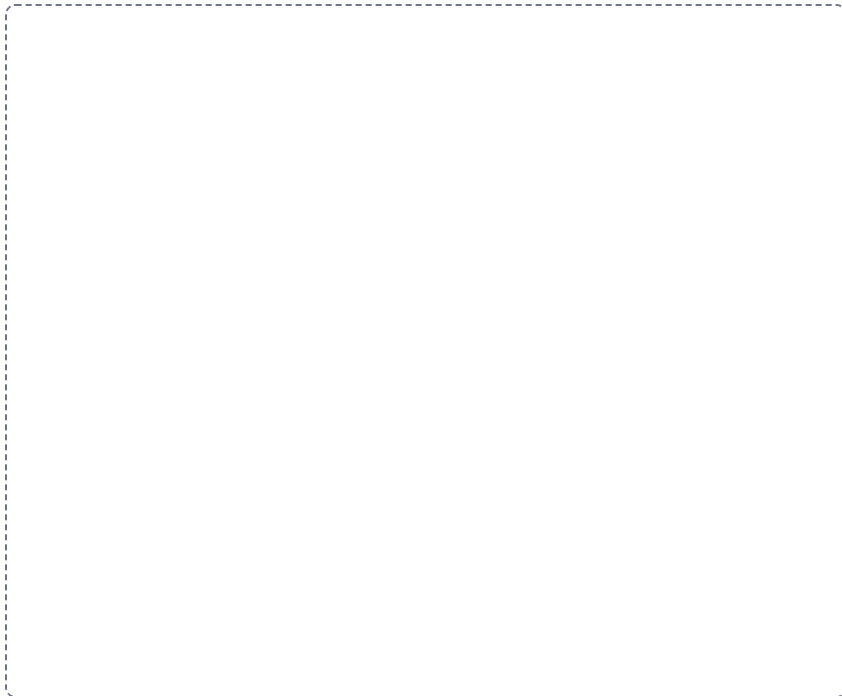
DRAWING 1

A synovial joint, sectioned

BRAIN DUMP FIRST

Two minutes. Every structure inside a synovial joint and what each one does.

Draw two bone ends meeting, with a gap between them. Cap each with articular cartilage. Draw the capsule wrapping the joint, with the fibrous membrane outside and the synovial membrane inside, and fill the cavity with synovial fluid. Add a meniscus between the surfaces, a fat pad, and a bursa outside the capsule. Label all of it.



In your notebook, head the page **M2-10 A synovial joint, sectioned** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name the four jobs of synovial fluid while pointing at where it sits.

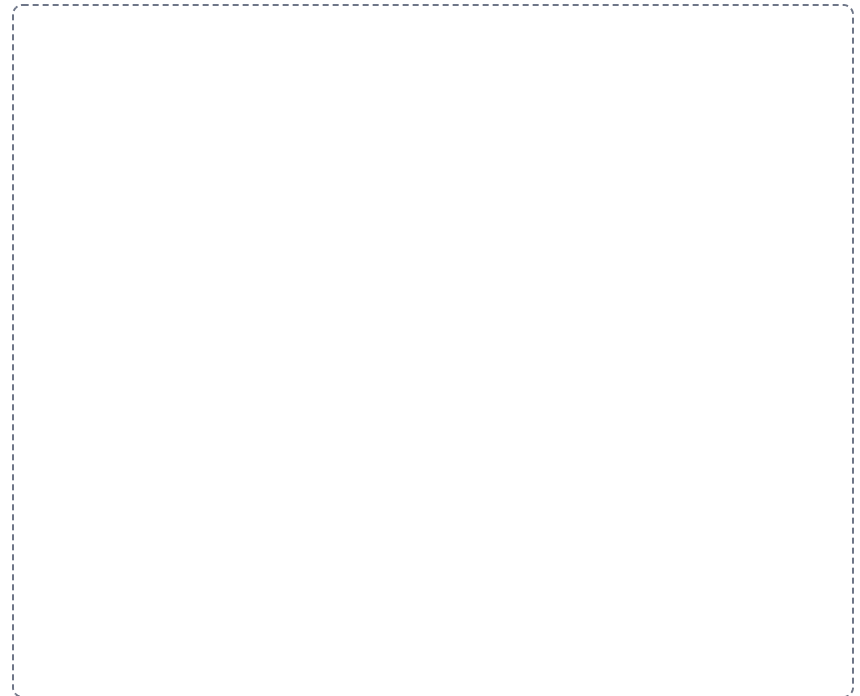
DRAWING 2

The knee, anterior and from above

BRAIN DUMP FIRST

Two minutes. Every ligament and both menisci.

Draw the knee from the front with the patella lifted away: femoral condyles, tibial plateau, both collateral ligaments outside, both cruciates crossing inside. Then draw the tibial plateau from above and put the C-shaped medial meniscus and the O-shaped lateral meniscus in place. Mark the three structures of the unhappy triad.



In your notebook, head the page **M2-11 The knee, anterior and from above** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain what stops the tibia sliding forward, and what happens when it fails.

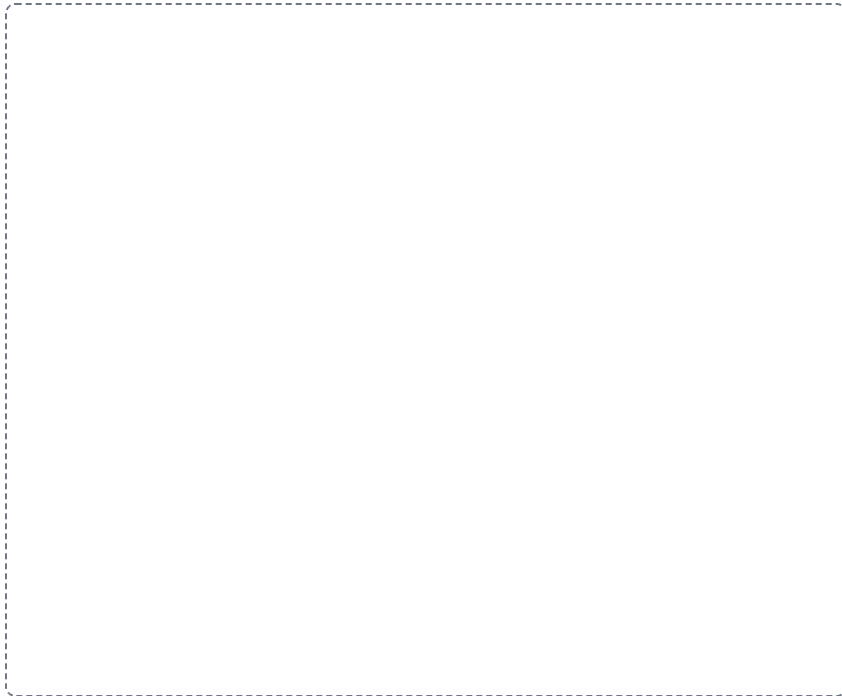
DRAWING 3

The shoulder, and why it dislocates

BRAIN DUMP FIRST

One minute. The glenohumeral joint and what actually holds it together.

Draw the humeral head against the glenoid cavity and mark how much larger the head is than the socket. Add the glenoid labrum deepening the rim, the loose capsule, and the coracohumeral and glenohumeral ligaments. Then draw the four rotator cuff muscles gripping the head, and label them.



In your notebook, head the page **M2-12 The shoulder, and why it dislocates** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say why the ligaments are not what stabilizes this joint, and name the four muscles that are.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

Classify the pubic symphysis by structure and by function, and name the tissue involved.

Answer. Structurally a cartilaginous joint, specifically a symphysis. Functionally an amphiarthrosis. The tissue is fibrocartilage.

Reasoning. Run the two classifications separately, then check they agree. Structure first: is there a synovial cavity? No, so it is fibrous or cartilaginous. What holds the bones? A broad flat disc of fibrocartilage, which is the definition of a symphysis. Function second: fibrocartilage allows a little give, so slightly movable, an amphiarthrosis. The two answers are consistent, which is the check that you classified it correctly.

Set A · Classify it

1. A tooth in its socket. Structural class _____ Functional class _____
2. The elbow. Synovial type _____ Axes _____
3. The epiphyseal plate. Structural class _____ Functional class _____
4. The carpometacarpal joint of the thumb. Synovial type _____ Axes _____
5. The atlantoaxial joint. Synovial type _____ Movement it produces _____

Set B · Name the movement

1. Turning the sole of the foot inward is _____.
2. Moving the thumb across the palm to touch the fingertips is _____.
3. Turning the palm to face posteriorly is _____.
4. Bending the head sideways toward the shoulder is _____.
5. Name the movements possible at a ball-and-socket joint.

Set C · Predict it

1. A lateral blow lands on a planted knee. Name the three structures likely torn and the name of the injury.

2. A patient has an unstable shoulder after an injury. Explain why the ligaments alone do not stabilize it, and name what does.

3. A knee swells immediately after an injury, and another swells the next day. Say what each timing suggests.

4. Distinguish osteoarthritis from rheumatoid arthritis by cause, joints affected, and symmetry.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

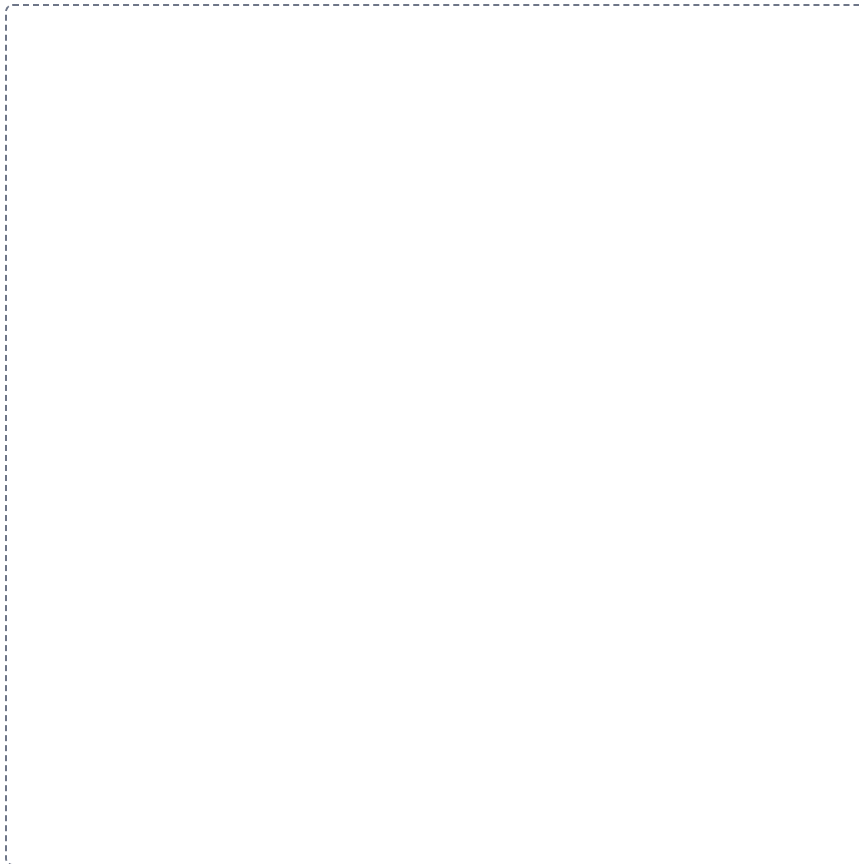
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M2-1

Structure crossed with function

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

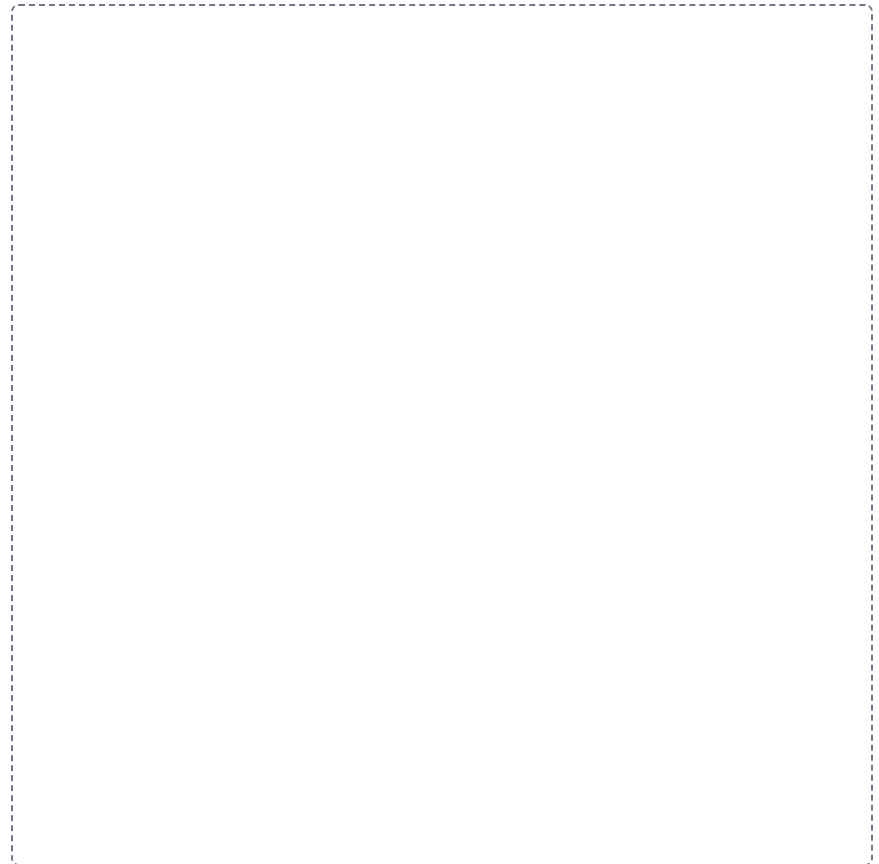


M2-2

Fibrous joints

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

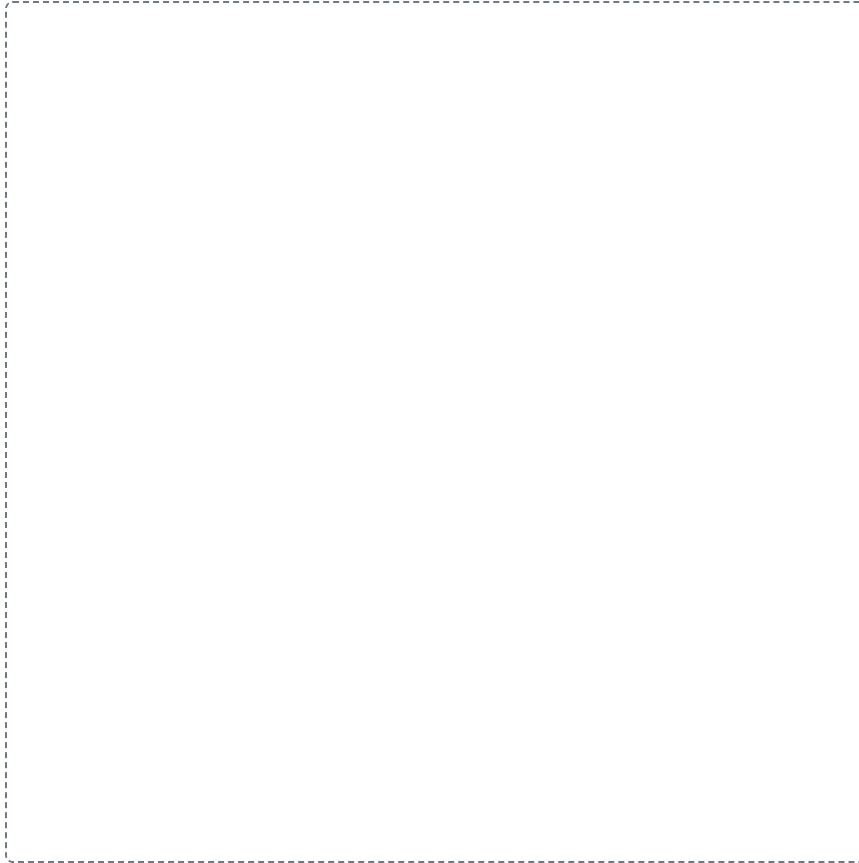


12-3

Synchondrosis against symphysis

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

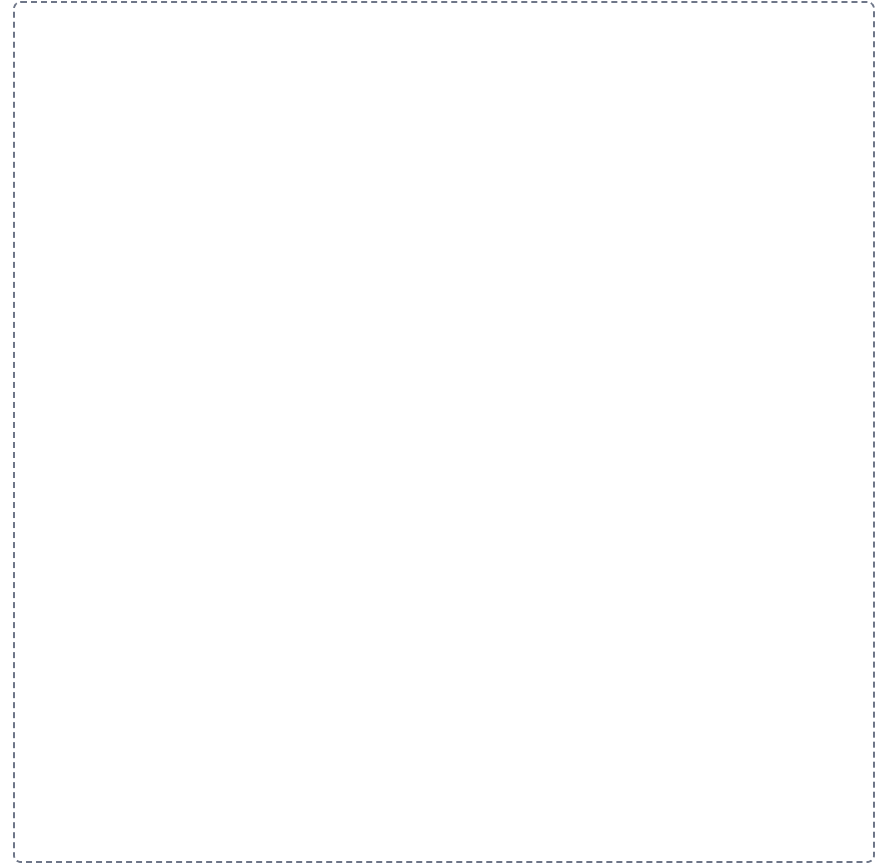


12-4

A synovial joint in section

HUB

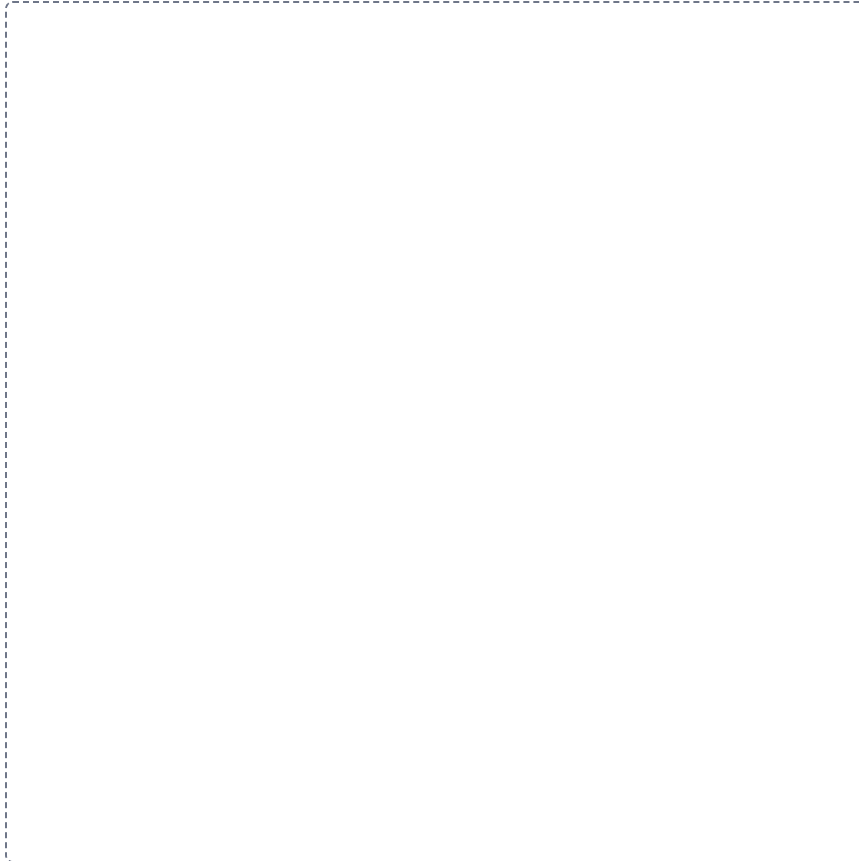
Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



The six synovial types

GRID

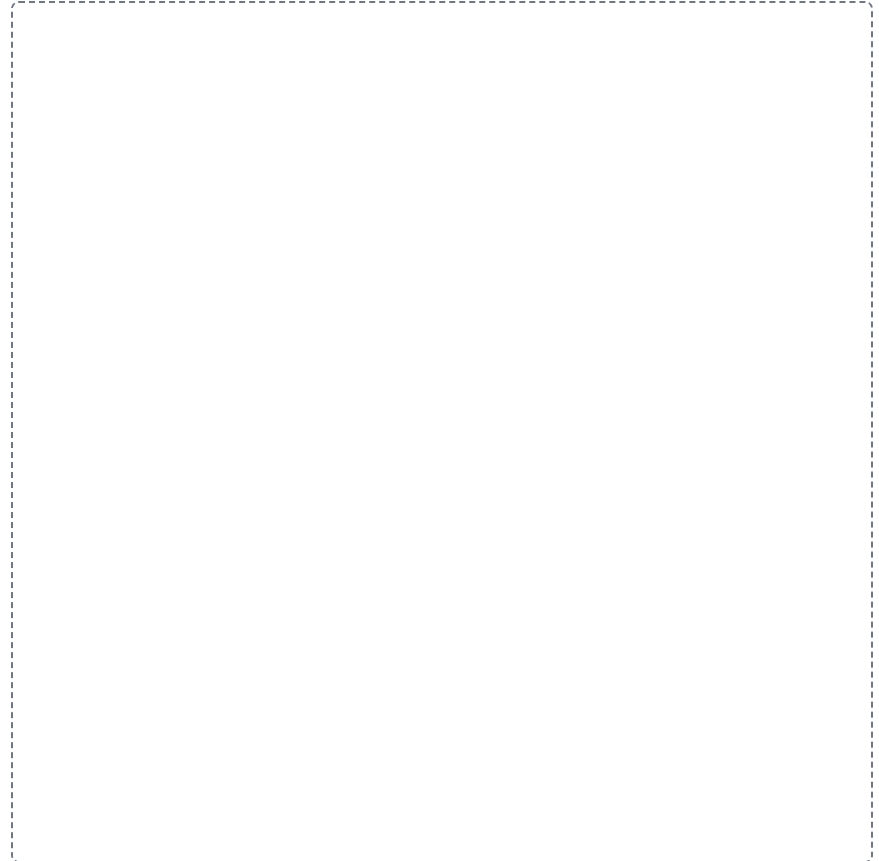
Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



The movements

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



Osteoarthritis against rheumatoid arthritis**SPLIT**

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

